

REALTEK

Quick Guide for wpa_supplicant Wi-Fi P2P test

Date: 2018/08/27

Version: 1.0

(1) Release note

Document Version	Note
V1.0	1. First release

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(2) P2P description:

Wi-Fi Direct (P2P) is a technology developed by Wi-Fi Alliance. It is a solution for Wi-Fi device-to-device connectivity. And it is also backward compatible with existing Wi-Fi Certified devices.

In usual, there are 3 stages in the Wi-Fi Direct scenario.

1. “Device Discovery”
2. “Group Formation” + “Provisioning” (WPS)
3. “Device connection” (DHCP)

The following picture will provide the overall concept for Wi-Fi Direct functionality and it will also contain these 3 stages described above.

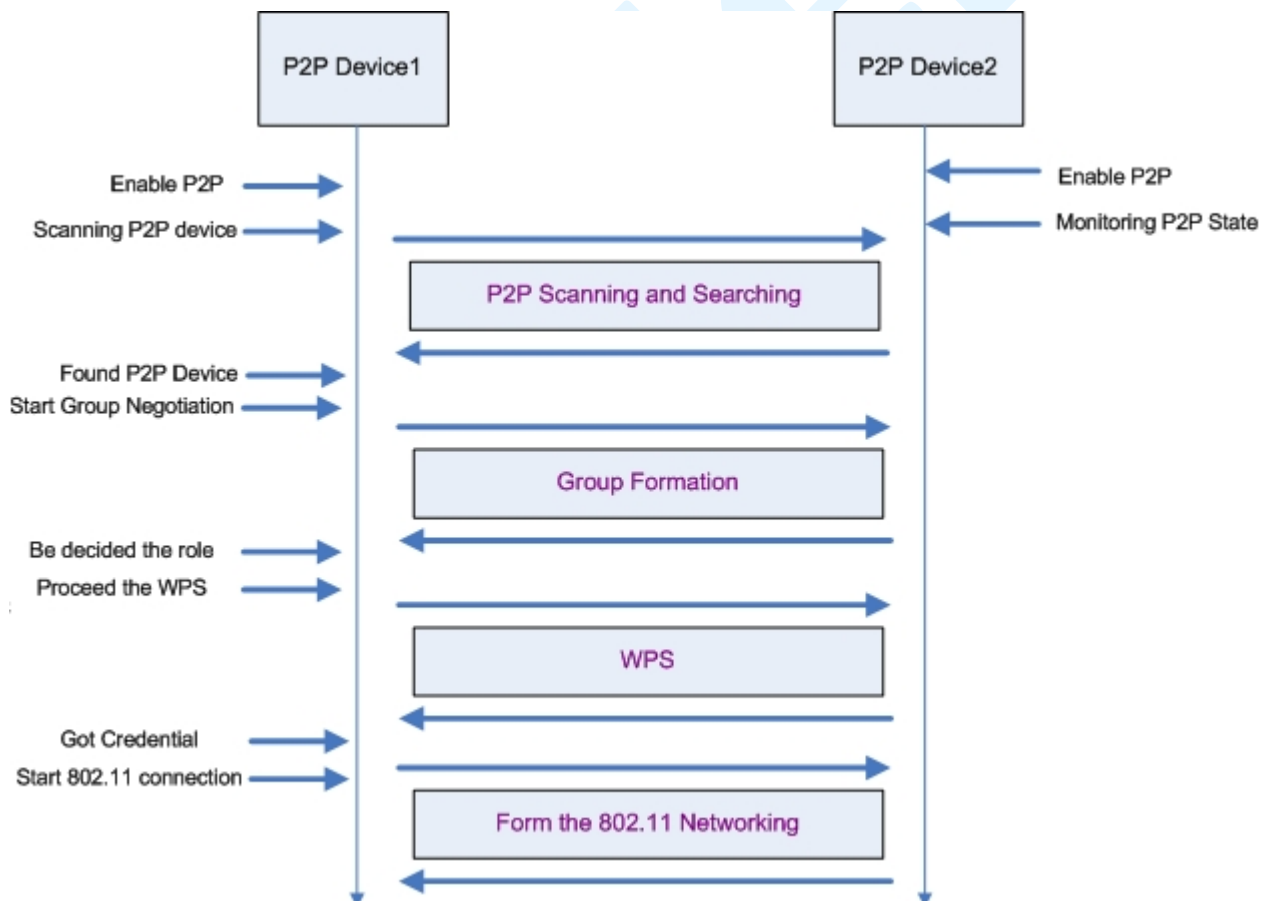


Figure1: Wi-Fi Direct Overview

The figure1 describes the basic Wi-Fi Direct scenario and this document will use this figure to do connection test.

(3) Support version:

- cfg80211 only

(4) How to start wpa_supplicant:

- Please reference another document under document folder. (wpa_cli_with_wpa_supplicant.pdf)

(5) P2P commands

Device Discovery

Command	Description
p2p_find [timeout in seconds]	Enables discovery – start sending probe request frames
p2p_stop_find	Stops discovery, or whatever you are doing (listen mode, connection process etc.)
p2p_listen [timeout in seconds]	Enable listen mode

Device Discovery : Discovered Peers List

Command	Description
p2p_peers [discovered]	Shows list of discovered peers (with 'discovered' – shows only fully discovered peers)
p2p_peer <P2P Device Address>	Show detailed information about discovered peers
p2p_flush	Flush p2p_state, and clears the discovered peer list

Group Formation

Command	Description
p2p_connect <peer device address> <pbcc/pin> [GO_intent=<0-15>] [auth/join]	GO_intent – initiate connection to another device (using entered group intent) Auth – WPS authorize incoming connection Join – connect to an existing GO No input – initiate connection using default GO intent
p2p_group_add [freq=<freq in MHz>]	Become an autonomous GO



(6) P2P Use case

Case 1 : Connect in pbc (Push button Control)

Case 1-1 : DUT #1 & DUT #2 use p2p_find to find out each other

Step #	DUT #1	DUT #2	Comments
1	p2p_find 30	p2p_find 30	Find p2p device
2	p2p_peers	p2p_peers	verify p2p candidates MAC ADDRESS
3	p2p_connect <DUT#2_MAC_ADDRESS> pbc go_intent=7		go_intent=7 means that there a same chance for both EVMs to become GO go_intent= 15 means that EVM will become GO go_intent= 0 means that EVM will become Client
4		p2p_connect <DUT#1_MAC_ADDRESS> pbc	

DUT_1_case_1-1

```
[root]# ./wpa_cli -i wlan0 status
wpa_state=DISCONNECTED
p2p_device_address=00:e0:4c:02:80:45
address=00:e0:4c:02:80:45

[root]# ./wpa_cli -i wlan0 p2p_find 30
OK

[root]# ./wpa_cli -i wlan0 p2p_peers
00:e0:4c:02:80:6e

[root]# ./wpa_cli -i wlan0 p2p_connect 00:e0:4c:02:80:6e pbc go_intent=7
OK

[root]# ./wpa_cli -i wlan0 status
bssid=00:e0:4c:02:80:6e
ssid=DIRECT-Cc
id=2
mode=station
pairwise_cipher=CCMP
group_cipher=CCMP
key_mgmt=WPA2-PSK
wpa_state=COMPLETED
ip_address=192.168.42.33
p2p_device_address=00:e0:4c:02:80:45
address=00:e0:4c:02:80:45
```

DUT_2_case_1-1

```
[root]# ./wpa_cli -i wlan0 status
wpa_state=DISCONNECTED
p2p_device_address=00:e0:4c:02:80:6e
address=00:e0:4c:02:80:6e

[root]# ./wpa_cli -i wlan0 p2p_find 30
OK

[root]# ./wpa_cli -i wlan0 p2p_peers
00:e0:4c:02:80:45

[root]# ./wpa_cli -i wlan0 p2p_connect 00:e0:4c:02:80:45 pbc
OK

[root]# ./wpa_cli status
bssid=00:e0:4c:02:80:6e
ssid=DIRECT-Cc
id=1
mode=P2P GO
pairwise_cipher=CCMP
group_cipher=CCMP
key_mgmt=WPA2-PSK
wpa_state=COMPLETED
ip_address=192.168.42.1
p2p_device_address=00:e0:4c:02:80:6e
address=00:e0:4c:02:80:6e
```

Case 1-2 : DUT #2 listen only DUT #1 use p2p_find to find out DUT #2

Step #	DUT #1	DUT #2	Comments
1	p2p_find 30	p2p_listen 60	Find p2p device
2	p2p_peers		verify p2p candidates MAC ADDRESS
3	p2p_connect <DUT#2_MAC_ADDRESS> pbc		go_intent=7 means that there a same chance for both EVMs to become GO go_intent= 15 means that EVM will become GO go_intent= 0 means that EVM will become Client
4		p2p_connect <DUT#1_MAC_ADDRESS> pbc	

DUT_1_case_1-2

```
[root]# ./wpa_cli -i wlan0 p2p_find 30
OK
```

```
[root]# ./wpa_cli -i wlan0 p2p_peers
00:e0:4c:02:80:6e
```

```
[root]# ./wpa_cli -i wlan0 p2p_connect 00:e0:4c:02:80:6e pbc
OK
```

```
[root]# ./wpa_cli -i wlan0 status
bssid=00:e0:4c:02:80:6e
ssid=DIRECT-5e
id=2
mode=station
pairwise_cipher=CCMP
group_cipher=CCMP
key_mgmt=WPA2-PSK
wpa_state=COMPLETED
ip_address=192.168.42.33
p2p_device_address=00:e0:4c:02:80:45
address=00:e0:4c:02:80:45
```

DUT_2_case_1-2

```
[root]# ./wpa_cli -i wlan0 p2p_listen 60
OK
```

```
[root]# ./wpa_cli -i wlan0 p2p_connect 00:e0:4c:02:80:45 pbc
OK
```

```
[root]# ./wpa_cli status
bssid=00:e0:4c:02:80:6e
ssid=DIRECT-5e
id=1
mode=P2P GO
pairwise_cipher=CCMP
group_cipher=CCMP
key_mgmt=WPA2-PSK
wpa_state=COMPLETED
ip_address=192.168.42.1
p2p_device_address=00:e0:4c:02:80:6e
address=00:e0:4c:02:80:6e
```


Case 2 : Connect in pbc (Push button Control) where DUT #1 is defined as the Auto Group Owner

Step #	DUT #1	DUT #2	Comments
1	p2p_find 30	p2p_find 30	
2	p2p_peers	p2p_peers	verify p2p candidates MAC ADDRESS
3	p2p_group_add		Define DUT #1 as GO
4	wps_pbc		
5		p2p_connect <DUT#1_MAC_ADDRESS> pbc join	

DUT_1_case_2

```
[root]# ./wpa_cli -i wlan0 p2p_find 30
OK
```

```
[root]# ./wpa_cli -i wlan0 p2p_peers
00:e0:4c:02:80:6e
```

```
[root]# ./wpa_cli -i wlan0 p2p_group_add
OK
```

```
[root]# ./wpa_cli -i wlan0 wps_pbc
```

DUT_2_case_2

```
[root]# ./wpa_cli -i wlan0 p2p_listen 60
OK
```

```
[root]# ./wpa_cli -i wlan0 p2p_connect 00:e0:4c:02:80:45 pbc join
OK
```

Case 3 : Connect in Pin (PIN Number) where DUT #1 is defined as the Auto Group Owner

Step # DUT #1 DUT #2

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Case 4 : Connection using PIN code

Step #	DUT #1	DUT #2	Comments
1	p2p_find 30	p2p_find 30	
2	p2p_peers	p2p_peers	verify p2p candidates MAC ADDRESS
3	p2p_connect <DUT#2_MAC_ADDRESS> pin auth		verify the <DUT#1_PIN_CODE>, will appear on the terminal
4		p2p_connect <DUT#1_MAC_ADDRESS> <DUT#1_PIN_CODE>	

DUT_1_case_4

```
[root]# ./wpa_cli -i wlan0 p2p_find 30
OK
```

```
[root]# ./wpa_cli -i wlan0 p2p_peers
00:e0:4c:02:80:6e
```

```
[root]# ./wpa_cli -i wlan0 p2p_connect 00:e0:4c:02:80:6e pin auth
53617170
```

DUT_2_case_4

```
[root]# ./wpa_cli -i wlan0 p2p_find 30
OK
```

```
[root]# ./wpa_cli -i wlan0 p2p_peers
00:e0:4c:02:80:45
```

```
[root]# ./wpa_cli -i wlan0 p2p_connect 00:e0:4c:02:80:45 53617170
OK
```